

UNIT 1 · TOPIC 1.5

Reservation Systems

The invisible engines that power every airline booking on Earth.

B.B.A. Aviation Management · Semester III · Unit 1: Overview of the Airline Industry
Air Ticketing & Fare Construction (2ACBAM32) · Arun Kumar

**BOARDING PASS**

LEARNER

STU

PROFESSIONAL

**PRO**

FLIGHT

ATF 1.5

GATE

PERIOD 5

DURATION

00:40

SEAT

SEM III

CLASS

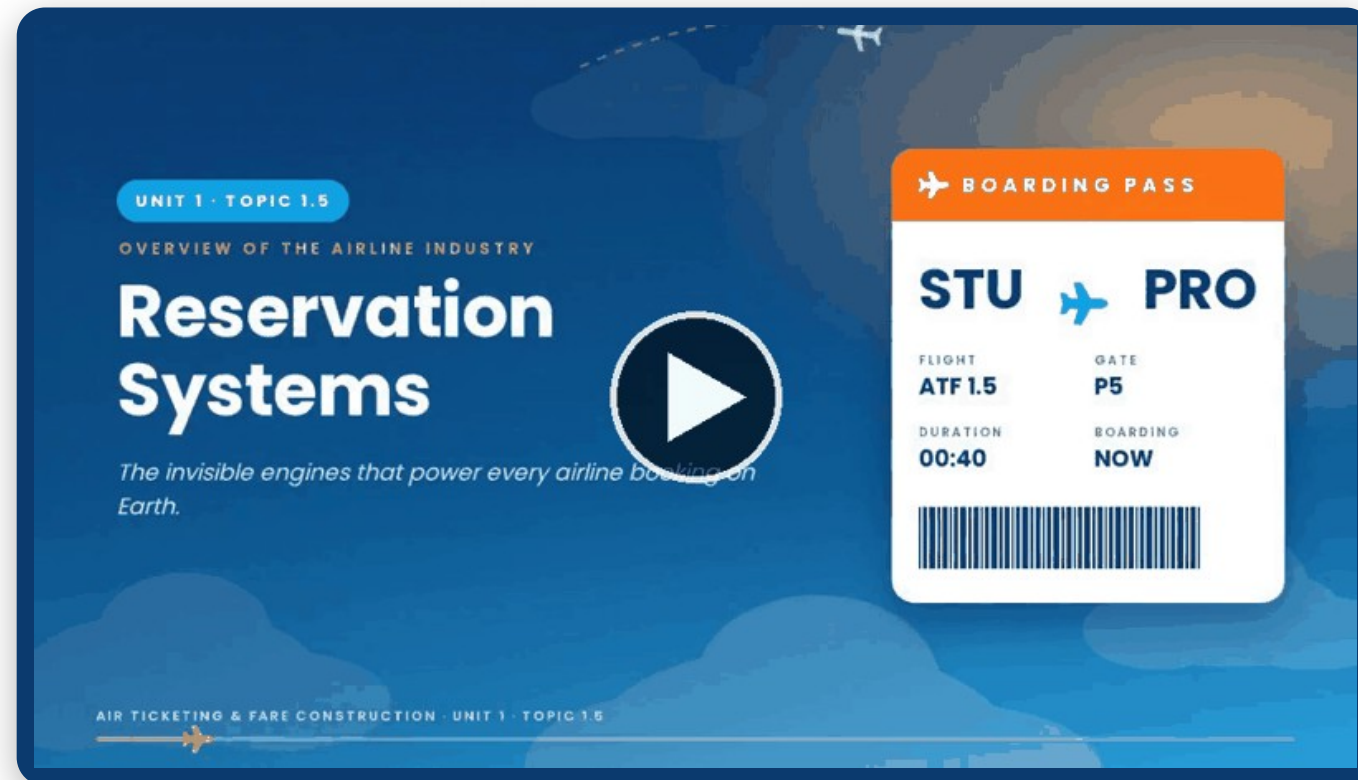
TECH-BASED

BOARDING

NOW

TOPIC 1.5

In-Flight Movie: Topic Intro



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02

TOPIC 1.5

Pre-Flight Question



THINK ABOUT IT

When you tap “Book” in an app, how many computer systems do you think talk to each other?

**THINK**

30 seconds alone

**PAIR**

1 minute with a neighbour

**SHARE**

2-3 answers with the class



TOPIC 1.5

Today's Flight Plan

1

1.5.1

Introduction to reservation systems: functions of a GDS

2

1.5.2

Major reservation systems: Amadeus, Sabre, Galileo and Travelport

3

1.5.3

Role of CRS and how they facilitate ticketing

4

1.5.4

System integration with airlines and OTAs: real-time information sharing

**BY LANDING, YOU WILL**

- Define GDS and CRS and their functions
- Compare the major GDS providers
- Trace how a booking moves between systems
- Explain APIs and NDC in modern distribution

Pedagogy: Technology-Based Learning

TOPIC 1.5.1

What Is a GDS?

A Global Distribution System is a marketplace connecting airlines to sellers worldwide.

**Central database**

Stores schedules, seat availability and fares from hundreds of airlines.

**Real-time links**

Connects to airline inventory so a sale updates seats instantly.

**One-stop shop**

Agents also book hotels, cars, rail and insurance in the same system.

**DEFINITION**

A GDS is a computerised network that lets travel agents and online sellers search, book and ticket travel products from many suppliers through one interface.



TOPIC 1.5.1

Core Functions of a GDS



Availability & Schedules

Search flights and seats across airlines in seconds.



Fare Quote & Pricing

Auto-price itineraries using filed fares and rules.



PNR Creation

Store the passenger record with a unique booking reference.



Ticketing & Reporting

Issue e-tickets and report sales to IATA BSP.



TOPIC 1.5.2

The Major GDS Providers



Amadeus

Founded 1987 by European airlines; HQ Madrid. Largest GDS by bookings; strong in Europe, Asia and India.



Sabre

Grew from American Airlines' 1960 SABRE system — the first airline computer reservation system. Strong in North America.



Travelport

Owns the Galileo, Apollo and Worldspan systems, now on the Travelport+ platform. Popular with Indian agencies.



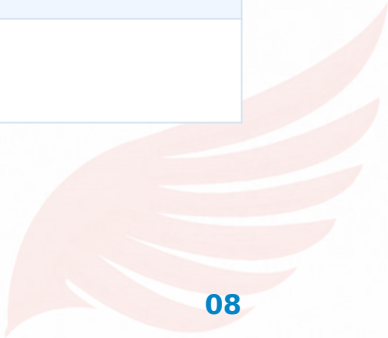
TOPIC 1.5.2

GDS Comparison at a Glance

Different command languages, same core purpose.

GDS	Origin	Availability command*	Retrieve PNR*
Amadeus	1987, Air France, Lufthansa, Iberia, SAS	AN15NOVDELBOM	RT
Sabre	1960, American Airlines + IBM	115NOVDELBOM	*ABC123
Galileo (Travelport)	1987, European airline group	A15NOVDELBOM	*ABC123
Worldspan (Travelport)	1990, Delta, Northwest, TWA	A15NOVDELBOM	*ABC123

*Note: *Simplified formats for teaching. Syntax varies by version and graphical interfaces now hide most commands.*



TOPIC 1.5.3

CRS vs GDS



CRS (Computer Reservation System)

- Airline's own inventory and reservation system
- Holds seats, fares and PNRs for that airline
- Also used for check-in and departure control
- Example: Amadeus Altéa, Navitaire (used by IndiGo)

VS



GDS (Global Distribution System)

- Neutral marketplace for many airlines
- Used by travel agents and OTAs
- Pulls data from each airline's CRS
- Example: Amadeus, Sabre, Travelport



TOPIC 1.5.3

How a CRS Makes Ticketing Possible



TOPIC 1.5.4

Integration with Airlines & OTAs

Information must flow in real time — or passengers get double-booked.

**APIs**

OTAs connect to GDSs and airlines through APIs to search and book instantly.

**IATA NDC**

New Distribution Capability: an XML standard letting airlines send rich offers (seats, bags, bundles) directly.

**Messaging**

Schedule changes and seat updates are pushed automatically to all connected systems.

**LOOK-TO-BOOK**

OTAs may run hundreds of searches for every booking. To save cost and speed up results, they cache fares — and must re-check live availability before payment.



TOPIC 1.5.4

Real-Time Data Journey of a Booking

Seconds after you click "Book"...



TOPIC 1.5

Cabin Crew Activity: GDS Simulator Challenge

**PEDAGOGY****Technology-Based Learning****TIME · 10 min****FORMAT · Individual / pairs**

- 1 Open a free GDS practice simulator or the instructor's screen share.
- 2 Run an availability search for DEL-BOM on a future date.
- 3 Identify: airline, flight number, departure time and open booking classes.
- 4 Compare the result with the same search on an airline website.



TOPIC 1.5*Click to reveal each answer*

Pre-Landing Check

Q1**Which GDS grew out of American Airlines' 1960 system?**

- A) Amadeus
- B) Sabre
- C) Galileo
- D) Worldspan

ANSWER**B****Q2****Galileo is today part of...**

- A) Amadeus
- B) Sabre
- C) Travelport
- D) IATA

ANSWER**C****Q3****An airline's own inventory system is called a...**

- A) GDS
- B) CRS / PSS
- C) OTA
- D) BSP

ANSWER**B****Q4****IATA's XML standard for richer airline offers is...**

- A) NDC
- B) PNR
- C) RBD
- D) ATPCO

ANSWER**A**

SAFE LANDING

Key Takeaways

- ✓ A GDS is a neutral marketplace linking airlines to agents and OTAs
- ✓ Big providers: Amadeus, Sabre and Travelport (Galileo, Worldspan)
- ✓ The airline CRS/PSS holds inventory; the GDS distributes it
- ✓ APIs and NDC enable real-time, richer offers

EXIT TICKET Write 1 thing you learned today + 1 question you still have.

NEXT DEPARTURE**Topic 1.6 — Passenger Name Record (PNR)**

Thank you · Questions?